Part A

A-15

a-) $d(4,5) = 2.11 \quad d(1,15) = 6.26 \quad d(4,1) = 4.39 \quad d(5,15) = 2.83$

$\Delta = 2.11 + 6.26 - 4.39 - 2.83 = 1.15$

$\Delta > 0 \rightarrow$ It is worsening

b-) $e^{\left(\frac{-1.15}{3}\right)} \approx 0.6816 = P$

c-) $U < P$ Therefore accept the move.

d-) $e^{\left(\frac{-1.15}{0.5}\right)} \approx 0.1$, Since $U > P$ reject the move.

Temperature represents the randomness / exploration level.
Lower temperature implies lower probability of acceptance.

A2)

a-)

Iteration	T
0	3
20	$2.7 = 3 * 0.9$
100	$1.77 = 3 * 0.9^5$
200	$1.05 = 3 * 0.9^{10}$

b-)

Current solution tracks the current position of the algorithm in the search space. And since, it is allowed to accept worsening solutions, tour length may increase from time to time.

However best-so-far only updated if current solution beats the best-so-far. In which case, current becomes the new best-so-far.